Invariants of Conic Sections

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Course project for MAT 7470, Wayne State University, Winter 2024, with Prof. Candelori.
Written up for this site in October 2026.

Move an ellipse in the plane, by turning it or sliding it, and its equation changes completely, yet it is the same curve. This project answers the natural question: which numbers, computed from the equation, do *not* change? Two such numbers turn out to be enough. Two conics with the same two numbers are the same curve in different positions.

1 The setting

A *conic* is the set of points (x, y) in the plane where a quadratic polynomial vanishes:

$$p(x,y) = ax^2 + 2bxy + cy^2 + 2dx + 2ey + f = 0, \qquad (a,b,c) \neq (0,0,0). \tag{1}$$

The factors 2 are there so that the coefficients fit into a symmetric matrix. With $v = (x, y, 1)^T$,

$$p(x,y) = v^T M v, \qquad M = \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix}, \qquad M_0 = \begin{pmatrix} a & b \\ b & c \end{pmatrix},$$

where M_0, the top-left corner of M, holds the quadratic terms alone.

> Calculation
>
> Multiply out:
>
> $$v^T M v = \begin{pmatrix} x & y & 1 \end{pmatrix} \begin{pmatrix} ax+by+d \\ bx+cy+e \\ dx+ey+f \end{pmatrix} = ax^2+bxy+dx+bxy+cy^2+ey+dx+ey+f = p(x,y).$$

Three numbers are made from the matrix:

$$T = a + c = \operatorname{tr} M_0, \qquad \delta = ac - b^2 = \det M_0, \qquad \Delta = \det M.$$

An equation up to a factor. The equations $p = 0$ and $\lambda p = 0$, for a number $\lambda \neq 0$, describe the same conic. So a number that truly belongs to the conic, and not to one particular equation of it, must not change when p is replaced by λp. Throughout, "a conic" means the equation $p = 0$ up to such a factor.[1]

[1] For a non-degenerate conic with real points this is the same as the set of points, since the set determines the equation up to a factor. For an equation like $x^2 + y^2 + 1 = 0$, with no real points at all, the equation carries more information than the (empty) set.

2 The main result

Theorem 1. *Suppose $\Delta \neq 0$. The two numbers*

$$f_1 = \frac{a+c}{\Delta^{1/3}} = \frac{T}{\Delta^{1/3}}, \qquad f_2 = \frac{ac-b^2}{\Delta^{2/3}} = \frac{\delta}{\Delta^{2/3}}$$

(real cube roots) do not change when the equation is multiplied by a nonzero number, nor when the conic is moved by a rigid motion: a rotation, a reflection, a translation, or any combination of them.

The proof has two independent parts:

- a rigid motion does not change T, δ or Δ at all (Section 3);
- multiplying p by λ multiplies T, δ, Δ by λ, λ^2, λ^3, and the powers of Δ in f_1, f_2 are exactly the ones that cancel these factors (Section 4).

Section 5 shows that f_1 and f_2 *do* change when the conic is enlarged, so they see the size of a conic, not only its shape. Sections 6 and 7 explain where Δ comes from and what the condition $\Delta \neq 0$ means, and Section 8 shows that f_1 and f_2 are enough to tell any two conics apart.

3 Rigid motions

A *rigid motion* of the plane is a map

$$\begin{pmatrix} x \\ y \end{pmatrix} \longmapsto Q \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_1 \\ t_2 \end{pmatrix}, \qquad Q^T Q = I.$$

The matrix Q is *orthogonal*: it keeps lengths and angles. The orthogonal 2×2 matrices are the rotations and the reflections,

$$Q_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \ (\det Q_\theta = 1), \qquad \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix} \ (\det = -1).$$

The rotation Q_θ is multiplication by the complex number $\cos\theta + i\sin\theta$, of absolute value 1, written as a matrix.

> Calculation
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> Write $\beta = \cos\theta$, $\gamma = \sin\theta$, so $\beta^2 + \gamma^2 = 1$. Then
>
> $$(\beta + i\gamma)(x + iy) = (\beta x - \gamma y) + i(\gamma x + \beta y), \quad \text{that is,} \quad \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} \beta & -\gamma \\ \gamma & \beta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$
>
> Multiplying by a complex number of absolute value 1 turns the plane without stretching it.

The key observation. With the extra coordinate 1, a rigid motion is a single matrix product $v \mapsto Gv$:

$$G = \begin{pmatrix} Q & t \\ 0\ 0 & 1 \end{pmatrix} = \begin{pmatrix} q_{11} & q_{12} & t_1 \\ q_{21} & q_{22} & t_2 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det G = \det Q = \pm 1.$$

Substituting the moved point into the equation gives

$$p(Gv) = (Gv)^T M(Gv) = v^T \big(G^T M G\big) v,$$

so **the new coefficient matrix is** $M' = G^T M G$. The matrix of the motion appears on the right, and its transpose on the left. (For a rotation the transpose is also the inverse, $Q_\theta^T = Q_{-\theta}$, which is why the left factor looks like the inverse rotation.)

Proposition 2. *A rigid motion does not change T, δ and Δ.*

Proof. For Δ: $\det M' = \det(G^T)\det M \det G = (\det G)^2 \Delta = \Delta$, because $\det G = \pm 1$.

For T and δ: the top-left corner of $G^T M G$ is $M_0' = Q^T M_0 Q$, whatever the translation t is. Since $Q^T = Q^{-1}$, the matrix $M_0' = Q^{-1} M_0 Q$ is similar to M_0, and similar matrices have the same trace and the same determinant. □

Calculation

Write M and G in blocks, with $w = (d, e)^T$:

$$M = \begin{pmatrix} M_0 & w \\ w^T & f \end{pmatrix}, \qquad G = \begin{pmatrix} Q & t \\ 0 & 1 \end{pmatrix}, \qquad G^T = \begin{pmatrix} Q^T & 0 \\ t^T & 1 \end{pmatrix}.$$

Then

$$MG = \begin{pmatrix} M_0 Q & M_0 t + w \\ w^T Q & w^T t + f \end{pmatrix}, \qquad G^T M G = \begin{pmatrix} Q^T M_0 Q & Q^T(M_0 t + w) \\ * & * \end{pmatrix}.$$

The corner contains no t: a translation never changes the quadratic terms.

The two basic motions, written out coefficient by coefficient, follow. The primes denote the coefficients of the new equation $p'(x, y) = p(Gv) = a'x^2 + 2b'xy + c'y^2 + 2d'x + 2e'y + f'$.

A rotation. Substituting $x \mapsto \beta x - \gamma y$, $y \mapsto \gamma x + \beta y$ ($\beta = \cos\theta$, $\gamma = \sin\theta$):

$$\begin{aligned} a' &= a\beta^2 + 2b\beta\gamma + c\gamma^2, & b' &= (c-a)\beta\gamma + b(\beta^2 - \gamma^2), & c' &= a\gamma^2 - 2b\beta\gamma + c\beta^2, \\ d' &= \beta d + \gamma e, & e' &= \beta e - \gamma d, & f' &= f. \end{aligned}$$

Calculation

Expand $p(\beta x - \gamma y,\ \gamma x + \beta y)$ and collect terms. The quadratic part:

$$\begin{aligned} a(\beta x - \gamma y)^2 &= a\big(\beta^2 x^2 - 2\beta\gamma\, xy + \gamma^2 y^2\big), \\ 2b(\beta x - \gamma y)(\gamma x + \beta y) &= 2b\big(\beta\gamma\, x^2 + (\beta^2 - \gamma^2)\, xy - \beta\gamma\, y^2\big), \\ c(\gamma x + \beta y)^2 &= c\big(\gamma^2 x^2 + 2\beta\gamma\, xy + \beta^2 y^2\big). \end{aligned}$$

The coefficient of x^2 is $a\beta^2 + 2b\beta\gamma + c\gamma^2$, of y^2 is $a\gamma^2 - 2b\beta\gamma + c\beta^2$, and of xy is $2\big[(c - a)\beta\gamma + b(\beta^2 - \gamma^2)\big]$. The linear part:

$$2d(\beta x - \gamma y) + 2e(\gamma x + \beta y) = 2(\beta d + \gamma e)\, x + 2(\beta e - \gamma d)\, y.$$

A direct check: $a' + c' = (a + c)(\beta^2 + \gamma^2) = a + c$. In matrix form the whole substitution is

$$\begin{pmatrix} a' & b' & d' \\ b' & c' & e' \\ d' & e' & f' \end{pmatrix} = \begin{pmatrix} \beta & \gamma & 0 \\ -\gamma & \beta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} \begin{pmatrix} \beta & -\gamma & 0 \\ \gamma & \beta & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

with both outer matrices of determinant $\beta^2 + \gamma^2 = 1$.

A translation. Substituting $x \mapsto x + t_1$, $y \mapsto y + t_2$ leaves a, b, c unchanged and gives

$$d' = at_1 + bt_2 + d, \qquad e' = bt_1 + ct_2 + e, \qquad f' = p(t_1, t_2).$$

Calculation

Expand:

$$\begin{aligned} p(x + t_1, y + t_2) &= a(x + t_1)^2 + 2b(x + t_1)(y + t_2) + c(y + t_2)^2 + 2d(x + t_1) + 2e(y + t_2) + f \\ &= ax^2 + 2bxy + cy^2 + 2(at_1 + bt_2 + d)\, x + 2(bt_1 + ct_2 + e)\, y + p(t_1, t_2). \end{aligned}$$

In matrix form,

$$\begin{pmatrix} a' & b' & d' \\ b' & c' & e' \\ d' & e' & f' \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_1 & t_2 & 1 \end{pmatrix} \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} \begin{pmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \\ 0 & 0 & 1 \end{pmatrix},$$

and both outer matrices have determinant 1, so $\Delta' = \Delta$. Since a, b, c do not change, neither do T and δ.

4 Rescaling the equation

Replacing p by λp replaces M by λM, so

$$T \mapsto \lambda T, \qquad \delta \mapsto \lambda^2 \delta, \qquad \Delta \mapsto \lambda^3 \Delta.$$

The reason is degree: T is of degree 1 in the coefficients, δ of degree 2, Δ of degree 3. Dividing T by $\Delta^{1/3}$ and δ by $\Delta^{2/3}$ gives expressions of degree 0, and these are exactly f_1 and f_2. This completes the proof of Theorem 1.

Calculation

$\det(\lambda M) = \lambda^3 \det M$ for a 3×3 matrix, and $\det(\lambda M_0) = \lambda^2 \det M_0$. Then

$$\frac{\lambda T}{(\lambda^3 \Delta)^{1/3}} = \frac{\lambda T}{\lambda\, \Delta^{1/3}} = f_1, \qquad \frac{\lambda^2 \delta}{(\lambda^3 \Delta)^{2/3}} = \frac{\lambda^2 \delta}{\lambda^2\, \Delta^{2/3}} = f_2.$$

The real cube root satisfies $(\lambda^3 \Delta)^{1/3} = \lambda\, \Delta^{1/3}$ also for negative λ, which is why real cube roots are used.

Remark (why Δ is needed). The quotient $T^2/\delta = (a+c)^2/(ac-b^2)$ is also unchanged by rigid motions and by rescaling, and it does not involve Δ. But it ignores the linear and constant terms completely: every circle gives the value 4, whatever its radius, and every ellipse $x^2/\alpha^2 + y^2/\beta^2 = 1$ gives $(\alpha^2+\beta^2)^2/(\alpha\beta)^2$, which depends only on the ratio α/β. It sees the shape of a conic and not its size. In fact $T^2/\delta = f_1^2/f_2$: it is the combination of f_1 and f_2 that forgets the size (see the next section).

5 Enlarging is not a rigid motion

Enlarging the plane by a factor $\mu \neq 0$, that is substituting $x \mapsto \mu x$, $y \mapsto \mu y$, multiplies the quadratic terms by μ^2, the linear terms by μ, and leaves the constant alone. Then

$$T \mapsto \mu^2 T, \qquad \delta \mapsto \mu^4 \delta, \qquad \Delta \mapsto \mu^4 \Delta, \qquad \text{so} \qquad f_1 \mapsto \mu^{2/3} f_1, \qquad f_2 \mapsto \mu^{4/3} f_2.$$

So f_1 and f_2 change (unless $\mu = \pm 1$, and $\mu = -1$ is the rotation by π). They measure size as well as shape. For example, for the circle $x^2 + y^2 = r^2$,

$$f_1 = -2\,r^{-2/3}, \qquad f_2 = r^{-4/3},$$

so circles of different radii have different values.

Calculation

The new matrix is SMS with $S = \operatorname{diag}(\mu, \mu, 1)$:

$$\begin{pmatrix} \mu^2 a & \mu^2 b & \mu d \\ \mu^2 b & \mu^2 c & \mu e \\ \mu d & \mu e & f \end{pmatrix} = \begin{pmatrix} \mu & 0 & 0 \\ 0 & \mu & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} \begin{pmatrix} \mu & 0 & 0 \\ 0 & \mu & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

so $\Delta \mapsto (\det S)^2 \Delta = \mu^4 \Delta$. Then

$$f_1 \mapsto \frac{\mu^2 T}{\mu^{4/3} \Delta^{1/3}} = \mu^{2/3} f_1, \qquad f_2 \mapsto \frac{\mu^4 \delta}{\mu^{8/3} \Delta^{2/3}} = \mu^{4/3} f_2.$$

For the circle, $p = x^2 + y^2 - r^2$, so $a = c = 1$, $b = d = e = 0$, $f = -r^2$, and $T = 2$, $\delta = 1$, $\Delta = -r^2$.

6 Where Δ comes from: the center

When $\delta \neq 0$, the conic has a *center*: the point (x_0, y_0) where both partial derivatives of p vanish,

$$ax_0 + by_0 + d = 0, \qquad bx_0 + cy_0 + e = 0, \qquad \text{so} \qquad x_0 = \frac{be - cd}{\delta}, \quad y_0 = \frac{bd - ae}{\delta}.$$

Translating the center to the origin removes the linear terms (these are the formulas for d', e' in Section 3), and the constant term becomes the value of p at the center:

Proposition 3. *If $\delta \neq 0$, then $p(x_0, y_0) = \dfrac{\Delta}{\delta}$, that is, $\Delta = \delta \cdot p(\text{center})$.*

This is where Δ comes from. A rotation then turns the axes of the conic to the coordinate axes (the spectral theorem: the symmetric matrix M_0 is $Q^T \operatorname{diag}(\lambda_1, \lambda_2) Q$ for a rotation Q), and the equation becomes

$$\lambda_1 x^2 + \lambda_2 y^2 + \frac{\Delta}{\delta} = 0, \qquad \lambda_1 + \lambda_2 = T, \quad \lambda_1 \lambda_2 = \delta, \tag{2}$$

where λ_1, λ_2 are the eigenvalues of M_0.

Calculation

Completing the square, first in x and then in y, shows the same thing when $a \neq 0$:

$$\begin{aligned} p(x,y) &= a\Big(x^2 + \tfrac{2b}{a}xy\Big) + cy^2 + 2dx + 2ey + f \\ &= a\Big(x + \tfrac{b}{a}y\Big)^2 + \tfrac{ac-b^2}{a}\,y^2 + 2d\Big(x + \tfrac{b}{a}y\Big) + 2\,\tfrac{ae-bd}{a}\,y + f \\ &= a\Big(x + \tfrac{b}{a}y + \tfrac{d}{a}\Big)^2 + \tfrac{\delta}{a}\,y^2 + 2\,\tfrac{ae-bd}{a}\,y + f - \tfrac{d^2}{a} \\ &= a\Big(x + \tfrac{b}{a}y + \tfrac{d}{a}\Big)^2 + \tfrac{\delta}{a}\Big(y + \tfrac{ae-bd}{\delta}\Big)^2 + f - \tfrac{d^2}{a} - \tfrac{(ae-bd)^2}{a\,\delta}. \end{aligned}$$

The constant at the end is Δ/δ: over the common denominator $a\delta$, its numerator is

$$af\delta - d^2\delta - (ae - bd)^2 = a\big(acf - ae^2 - b^2 f + 2bde - cd^2\big) = a\,\Delta,$$

using the expansion $\Delta = acf - ae^2 - b^2 f + 2bde - cd^2$ of the 3×3 determinant. Both squares vanish exactly at the center, so the constant is $p(x_0, y_0)$.

The parabolic case $\delta = 0$. Then $T \neq 0$ (if $ac = b^2$ and $a + c = 0$, then $a = b = c = 0$), the eigenvalues of M_0 are 0 and T, and there is no center. If moreover $\Delta \neq 0$, a rotation and a translation bring the equation to

$$Ty^2 + 2kx = 0, \qquad k^2 = -\frac{\Delta}{T}. \tag{3}$$

Calculation

After a rotation, $M_0 = \operatorname{diag}(0, T)$, so $p = Ty^2 + 2d'x + 2e'y + f'$, and

$$\Delta = \det \begin{pmatrix} 0 & 0 & d' \\ 0 & T & e' \\ d' & e' & f' \end{pmatrix} = -Td'^2.$$

So $\Delta \neq 0$ means $d' \neq 0$. Completing the square in y gives $T(y + e'/T)^2 + 2d'x + (\text{constant})$, and a translation in x and in y removes both the y term and the constant. The determinant is unchanged, so $k = d'$ satisfies $\Delta = -Tk^2$.

7 The meaning of $\Delta = 0$

Proposition 4. *If p is a product of two linear polynomials, so that the conic is a pair of lines, then $\Delta = 0$.*

Proof. Write $p(x,y) = (u \cdot v)(w \cdot v)$ with $v = (x,y,1)^T$ and two fixed vectors u, w. Then $p = v^T u\, w^T v = v^T\big(\frac{1}{2}(uw^T + wu^T)\big)v$, so $M = \frac{1}{2}(uw^T + wu^T)$. Every column of this matrix is a combination of u and w, so its rank is at most 2 and $\det M = 0$. □

Calculation

With $u = (\alpha_1, \beta_1, \gamma_1)$ and $w = (\alpha_2, \beta_2, \gamma_2)$, expanding $(\alpha_1 x + \beta_1 y + \gamma_1)(\alpha_2 x + \beta_2 y + \gamma_2)$ gives

$$a = \alpha_1\alpha_2, \quad 2b = \alpha_1\beta_2 + \alpha_2\beta_1, \quad c = \beta_1\beta_2, \quad 2d = \alpha_1\gamma_2 + \alpha_2\gamma_1, \quad 2e = \beta_1\gamma_2 + \beta_2\gamma_1, \quad f = \gamma_1\gamma_2,$$

which are exactly the entries of $\frac{1}{2}(uw^T + wu^T)$:

$$M = \frac{1}{2}\begin{pmatrix} 2\alpha_1\alpha_2 & \alpha_1\beta_2 + \alpha_2\beta_1 & \alpha_1\gamma_2 + \alpha_2\gamma_1 \\ \alpha_1\beta_2 + \alpha_2\beta_1 & 2\beta_1\beta_2 & \beta_1\gamma_2 + \beta_2\gamma_1 \\ \alpha_1\gamma_2 + \alpha_2\gamma_1 & \beta_1\gamma_2 + \beta_2\gamma_1 & 2\gamma_1\gamma_2 \end{pmatrix}.$$

The converse also holds (a standard fact from linear algebra, not proved here): $\Delta = 0$ exactly when p is a product of two linear polynomials, possibly with complex coefficients. So the condition $\Delta \neq 0$ in Theorem 1 says that the conic is *non-degenerate*: a genuine ellipse, hyperbola or parabola, not a pair of lines or a single point.

Can a rigid motion remove the linear and constant terms? This question was raised in class. The answer is no when $\Delta \neq 0$. Without those terms the equation would be $ax^2 + 2bxy + cy^2 = 0$, whose matrix has last row $(0,0,0)$ and so $\Delta = 0$; but a rigid motion does not change Δ. More precisely:

- if $\delta \neq 0$, the linear terms can be removed (move the center to the origin), but the constant term is then $\Delta/\delta \neq 0$, by Proposition 3;
- if $\delta = 0$, not even all the linear terms can be removed: one remains, as in (3).

The substitution $x \mapsto x + \frac{b}{a}y$ used when completing the square does remove the xy term, but it is a shear, not a rigid motion: it changes the shape of the curve.

8 Two numbers are enough

Theorem 5. *Two conics with $\Delta \neq 0$ can be moved onto each other by a rigid motion if and only if they have the same values of f_1 and f_2.*

One direction is Theorem 1. For the other, the values of f_1, f_2 determine the normal form (2) or (3), and two conics with the same normal form are moved onto each other by going through it.

> Calculation
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> *Central case* ($\delta \neq 0$, equivalently $f_2 \neq 0$). Multiply the equation so that the constant term Δ/δ in (2) equals -1; then $\Delta = -\delta = -\lambda_1\lambda_2$, and
>
> $$f_2 = \frac{\lambda_1\lambda_2}{(\lambda_1\lambda_2)^{2/3}} = (\lambda_1\lambda_2)^{1/3}, \qquad f_1 = \frac{\lambda_1+\lambda_2}{(-\lambda_1\lambda_2)^{1/3}} = -\frac{\lambda_1+\lambda_2}{f_2}.$$
>
> So $\lambda_1\lambda_2 = f_2^3$ and $\lambda_1 + \lambda_2 = -f_1 f_2$: the numbers λ_1, λ_2 are the two roots of
>
> $$t^2 + f_1 f_2\, t + f_2^3 = 0,$$
>
> determined up to their order, and exchanging them is a rotation by $90°$.
> *Parabolic case* ($\delta = 0$, equivalently $f_2 = 0$). Multiply the equation so that $T = 1$ in (3); then $\Delta = -k^2$ and $f_1 = (-k^2)^{-1/3} = -|k|^{-2/3}$, which determines $|k|$. The sign of k is changed by the rotation by π.

The type of a conic. Since $\Delta^{2/3} > 0$, the sign of f_2 is the sign of δ, and the normal forms show what it means. For $\Delta \neq 0$:

- $f_2 > 0$ and $f_1 < 0$: an ellipse;
- $f_2 > 0$ and $f_1 > 0$: no real points (like $x^2 + y^2 + 1 = 0$);
- $f_2 < 0$: a hyperbola;
- $f_2 = 0$: a parabola (then always $f_1 < 0$).

For the standard equations, with $\alpha, \beta, q > 0$:

$$\begin{aligned}
&\text{ellipse } \frac{x^2}{\alpha^2} + \frac{y^2}{\beta^2} = 1: && f_1 = -\frac{\alpha^2+\beta^2}{(\alpha\beta)^{4/3}}, && f_2 = (\alpha\beta)^{-2/3},\\
&\text{hyperbola } \frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1: && f_1 = \frac{\beta^2-\alpha^2}{(\alpha\beta)^{4/3}}, && f_2 = -(\alpha\beta)^{-2/3},\\
&\text{parabola } y^2 = 4qx: && f_1 = -(4q^2)^{-1/3}, && f_2 = 0.
\end{aligned}$$

For the ellipse, f_2 gives the product $\alpha\beta$ and then f_1 gives $\alpha^2 + \beta^2$, so the two half-axes are recovered, as Theorem 5 promises.

> Calculation
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> *Type.* In (2) with $\delta = \lambda_1\lambda_2 > 0$, both λ_i have the sign of T, and there are real points exactly when the constant Δ/δ has the opposite sign, that is, when $T\Delta < 0$, that is, when $f_1 < 0$. With $\delta < 0$ the λ_i have opposite signs: a hyperbola. In the parabolic case $T\Delta = -T^2k^2 < 0$.
> *Ellipse.* Multiply by $\alpha^2\beta^2$: $\beta^2x^2 + \alpha^2y^2 - \alpha^2\beta^2 = 0$, so $T = \alpha^2 + \beta^2$, $\delta = \alpha^2\beta^2$, $\Delta = -\alpha^4\beta^4$.
> *Hyperbola.* $\beta^2x^2 - \alpha^2y^2 - \alpha^2\beta^2 = 0$, so $T = \beta^2 - \alpha^2$, $\delta = -\alpha^2\beta^2$, $\Delta = \alpha^4\beta^4$.
> *Parabola.* $y^2 - 4qx = 0$, so $c = 1$, $d = -2q$, all others 0: $T = 1$, $\delta = 0$, $\Delta = -4q^2$.

9 The geometric definitions agree

The sign of $\delta = ac - b^2$ can also be read off from the classical definitions of the three curves by distances.

Ellipse and hyperbola. Let F_1, F_2 be two points (the foci) at distance $2m$ from each other, and d_1, d_2 the distances from a point $P = (x, y)$ to them. The ellipse is where $d_1 + d_2 = L$, and the hyperbola where $|d_1 - d_2| = L$, for a fixed length L. Squaring twice turns either condition into the same polynomial equation,

$$L^4 - 2L^2\big(d_1^2 + d_2^2\big) + \big(d_1^2 - d_2^2\big)^2 = 0,$$

which is quadratic in x, y, with

$$\delta = ac - b^2 = 16L^2\big(L^2 - 4m^2\big).$$

For the ellipse $L = 2\alpha$ and $\alpha^2 = \beta^2 + m^2$, so $L^2 - 4m^2 = 4\beta^2 > 0$. For the hyperbola $L = 2\alpha$ and $m^2 = \alpha^2 + \beta^2$, so $L^2 - 4m^2 = -4\beta^2 < 0$. (Here α, β are the half-axes, as in the standard equations above.)

> Calculation
>
> *Squaring.* From $d_1 \pm d_2 = \pm L$: $d_1^2 + d_2^2 \pm 2d_1d_2 = L^2$, so $\pm 2d_1d_2 = L^2 - (d_1^2 + d_2^2)$. Squaring again,
>
> $$4d_1^2d_2^2 = L^4 - 2L^2(d_1^2 + d_2^2) + (d_1^2 + d_2^2)^2,$$
>
> and $(d_1^2 + d_2^2)^2 - 4d_1^2d_2^2 = (d_1^2 - d_2^2)^2$ gives the equation above. Every sign choice gives the same equation, which is why the ellipse and the hyperbola lead to the same formula.
> *The coefficients.* With $F_i = (h_i, k_i)$, $u = h_2 - h_1$, $w = k_2 - k_1$ (so $u^2 + w^2 = 4m^2$):
>
> $$d_1^2 + d_2^2 = 2x^2 + 2y^2 + (\text{linear}), \qquad d_1^2 - d_2^2 = 2ux + 2wy + (\text{constant}).$$
>
> So the quadratic part of the equation is $-4L^2(x^2 + y^2) + 4(ux + wy)^2$, that is,
>
> $$a = 4u^2 - 4L^2, \qquad b = 4uw, \qquad c = 4w^2 - 4L^2,$$
>
> and
>
> $$ac - b^2 = 16\big[(u^2 - L^2)(w^2 - L^2) - u^2w^2\big] = 16\big[L^4 - L^2(u^2 + w^2)\big] = 16L^2(L^2 - 4m^2).$$

Parabola. The parabola with focus (h, k) and directrix the line $rx + ty + s = 0$ is the set of points at equal distance from both:

$$\sqrt{(x - h)^2 + (y - k)^2} = \frac{|rx + ty + s|}{\sqrt{r^2 + t^2}}.$$

Its quadratic part is $t^2x^2 - 2rt\,xy + r^2y^2$, so $a = t^2$, $b = -rt$, $c = r^2$, and $\delta = r^2t^2 - r^2t^2 = 0$.

Calculation

Square and multiply by $r^2 + t^2$:

$$(r^2 + t^2)\big((x-h)^2 + (y-k)^2\big) - (rx + ty + s)^2 = 0.$$

The quadratic terms are $(r^2 + t^2)(x^2 + y^2) - (r^2x^2 + 2rt\,xy + t^2y^2) = t^2x^2 - 2rt\,xy + r^2y^2$.